



INDRA KUMAR

Applied AI Engineer | Generative AI | RAG | LLM Fine-Tuning
Indian Institute of Science, Bengaluru

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Professional Summary

Applied AI Engineer with an M.Tech in Robotics and Autonomous Systems from IISc, currently working in Product Development & R&D at OLA. Experienced in Retrieval-Augmented Generation, LLM fine-tuning, parameter-efficient adaptation, predictive maintenance, and production ML pipelines using Python, PyTorch, Hugging Face Transformers, FastAPI, Docker, and Azure.

Technical Skills

Programming & Data GenAI	Python, SQL, C++, Pandas, NumPy, scikit-learn, XGBoost Retrieval-Augmented Generation (RAG), LoRA, QLoRA, PEFT, Hugging Face Transformers, embeddings, semantic search, vector search, hybrid retrieval
ML & Deep Learning	PyTorch, TensorFlow, classification, regression, anomaly detection, CNNs, Vision Transformers, U-Net, reinforcement learning
APIs & Cloud	FastAPI, REST APIs, MLflow, Docker, Git, GitHub Actions, Azure AI Search, Azure Blob Storage, Azure Container Apps
Optimization & Robotics	Gurobi, Google OR-Tools, A* path planning, PID control, autonomous navigation

Professional Experience

OLA <i>Product Development and R&D</i>	Jun 2026 – Present <i>Bengaluru, India</i>
- Built an end-to-end RAG pipeline for industrial-equipment manuals using PyMuPDF, Azure Blob Storage, and Azure AI Search, covering PDF extraction, semantic chunking, embeddings, and incremental indexing. - Implemented keyword, dense-vector, and hybrid retrieval with semantic reranking; evaluated Recall@5, MRR, grounded-answer accuracy, and citation accuracy, and fine-tuned an instruction-following model using QLoRA/PEFT with page-level citations, confidence-based abstention, and hallucination guardrails. - Developed containerized FastAPI services for Azure Container Apps and contributed to Project Safai, modernizing internal platforms and reducing Salesforce dependency.	
Transportation Networks Laboratory, IISc <i>Graduate Student Researcher – M.Tech Thesis and EV Research Projects</i>	Sept 2024 – May 2026 <i>Bengaluru, India</i>
- Under Prof. Tarun Rambha, developed a two-stage stochastic mixed-integer Gurobi model jointly optimizing charger retrofits, charging schedules, request acceptance, and dynamic parking/energy pricing under demand uncertainty; evaluated 180- and 363-scenario datasets from ACN-Caltech sessions, California electricity prices, and charger-cost data, identifying profit-optimal conversion levels of 5–14% for a 100-space facility. - Under Prof. Prasant Msra (TCS Distinguished Senior Scientist, TCS Research), formulated EV routing with last-mile delivery and grid-energy discharge under time-window, battery, payload, station, distance, and fleet constraints; developed constraint-aware Maskable PPO and Linear Q-learning and benchmarked extended 100-customer Solomon instances across cost, fleet usage, runtime, grid discharge, and price sensitivity.	
Math Expert, Photomath <i>Subject Matter Expert</i>	Jan 2022 – May 2024 <i>Remote / Part-time</i>
Produced rigorous step-by-step LaTeX solutions in Linear Algebra, Calculus, Probability, and Optimization, and reviewed peer submissions for mathematical correctness, rigor, pedagogy, formatting, and adherence to quality rubrics.	
Selected Projects	
Multimodal Brain Tumor Segmentation with LLM-Adapted Transformers [Project Link]	Feb 2025 – May 2025
- Built 2D/3D BraTS 2021 segmentation pipelines for WT, TC, and ET from FLAIR, T1, T1CE, and T2 MRI; adapted Swin-UNETR and ViT-Small using LoRA on QKV/projection layers with frozen transformer weights and Dice/BCE loss. - Compared slice-based, tumor-centered 3D patch, and full-volume strategies with five-fold cross-validation; achieved 0.857 mean Dice with patches and 0.901 WT Dice with resizing using only 0.67–0.82M trainable parameters in the 2D variants.	
Path Planning and Control for Wheeled Robots Using Aerial Imagery [Project Link]	Sept 2025 – Oct 2025
- Built a TurtleBot4 navigation pipeline combining TensorFlow/Keras U-Net road segmentation, grid-based A* planning, PID control, and ROS; generated binary traversability maps using a four-level encoder-decoder with skip connections, batch normalization, and 0.1 dropout. - Extended A* to eight-neighbor search with diagonal costs and octile heuristic for smoother paths; tuned linear/angular PID controllers using cross-track and heading errors and demonstrated navigation on a physical TurtleBot4.	
Intelligent Automated OTS Scheduler – NPCI Hackathon Winner	Jan 2026
Built an autonomous operating-theatre scheduling system for hospital resource allocation, surgeon scheduling, equipment constraints, emergency cases, cancellations, and real-time rescheduling.	

